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November 18, 1977
SUPERSEDING
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May 24, 1962

FEDERAL SPECIFICATION

FURNACE, DENTAL LABORATORY, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electrically operated, muffle type, burn-out and heat-treating furnaces, with pyrometer, and automatic temperature controls suitable for use in dental laboratory operations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment.
Fed. Std. No. 151 - Metal, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6520

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies.)

Military Specifications.

MIL-L-10547 - Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.

Military Standards.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, Standards, and Drawings, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposals shall apply.

Underwriters' Laboratories (UL), Inc.

UL-1262 - Standard for Safety, Laboratory Equipment.

(Applications for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L. I. NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

American National Standards Institute (ANSI).

C73.11 - Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials Standards:

ASTM B193	- Electrical Resistivity.
ASTM C93	- Crushing Strength and Modulus of Rupture of Insulating Firebrick at Room Temperature.
ASTM C182	- Thermal Conductivity of Insulating Firebrick.
ASTM C202	- Thermal Conductivity of Refractory Brick.
ASTM C372	- Linear Thermal Expansion of Fired Whiteware Products by the Dilatometer Method.
ASTM C408	- Thermal Conductivity of Whiteware Ceramics.
ASTM C573	- Chemical Analysis of Fireclay and High-Alumina Refractories.
ASTM C674	- Flexural Properties of Ceramic Whiteware Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. All metal parts that are exposed to operating temperatures shall be fabricated from heat-resistant materials. Dissimilar metals in contact with each other shall be of such selection, or properly finished, to minimize electrolytic corrosion in the presence of moisture and high temperatures. Materials not specifically designated shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended.

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standard for Safety, Laboratory Equipment No. 1262. Proof shall consist of one of the following:

(a) UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories Inc. forward a copy of the UL report directly to the contracting Officer, upon the latter's request.

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(b) Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey of this laboratory. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts lists, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specification.

3.3 Design and construction. The furnace shall be an electrically operated muffle type for burn-out and heat treating in prosthodontic procedures (see figure 1). The housing shall be made in two sections, the upper or muffle section and the lower or instrument section (3.3.1). The two sections shall be insulated from each other by means of a full size integral insulating sheet (3.3.4.3). The muffle shall be five sided and molded in one piece of fused alumina refractory material (3.3.4.1). The muffle or upper section shall be equipped with a flue for conveying the hot gases to the outer air. The heating element wire shall be secured to the outside five surfaces of the muffle using refractory cement. The heating element wire (3.3.3) shall be completely covered by the refractory cement. The muffle and heating element assembly shall be easily replaceable as a unit. The five outside surfaces of the muffle shall be backed with a minimum of 1 1/2 inches of insulating firebrick or refractory fiberboard (3.3.4.1.1) to insulate against excessive loss of heat. The furnace shall be equipped with a counterbalanced door that opens downward (3.3.2). The door shall be equipped with a minimum of two hinges and two door stops positioned to hold the door in a horizontal position when fully open. The door hinges and door stops shall be of such design and structural strength as to properly support the weight of the door assembly. The door hinges and door stops shall be securely attached to the door and door frame by means of machine bolts. The counterweight arm and counterweight shall be so designed as to counterbalance the weight of the door in such a manner that the door can be opened or closed from either extreme position by exerting not more than two pounds of force on the counterweight arm when tested as specified in 4.4.2. The door shall remain in the extreme open position into which it is moved, presenting a level shelf in the open position, and rest flush against the door frame in the extreme closed position. The inner face of the door shall be faced with a refractory material (3.3.4.2) having a minimum thickness of 1/2 inch and an intermediate layer of insulating material (3.3.4.2.1) having a minimum thickness of 1 1/2 inch. The refractory and insulating components of the door shall be attached with a minimum of four (4) stainless steel bolts. A U-shaped trough, fabricated from 0.0478 inch thick (No. 18 Mfrs. Std.

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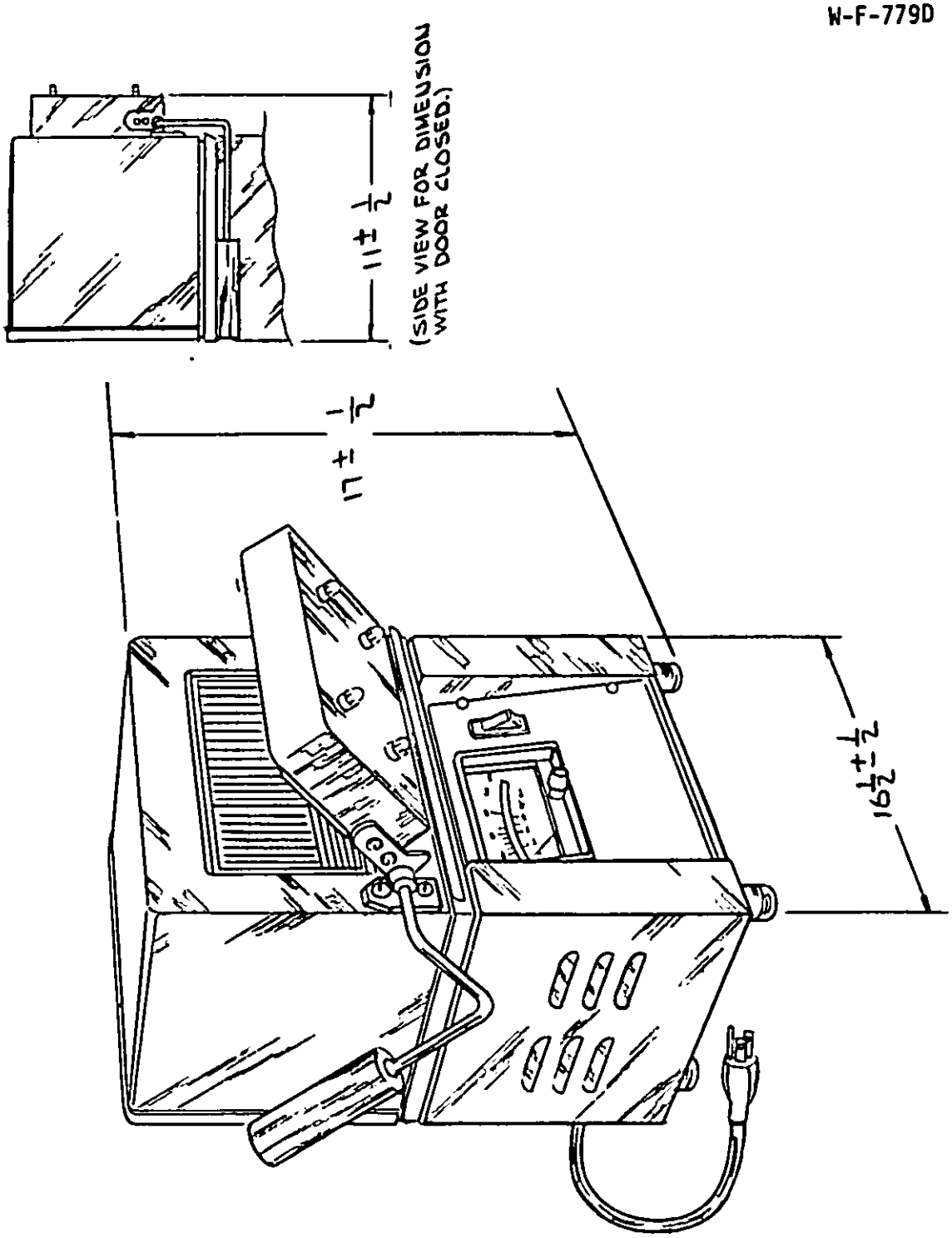


FIGURE 1. Dental laboratory furnace

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Gage), minimum, cold rolled steel, shall be securely attached to the bottom front surface of the upper or muffle section of the housing and shall be mounted flush with the front of the door frame surface. The trough shall be 5/16 ± 1/16 inch wide by 3/8 ± 1/16 inch deep, open at both ends, and shall extend the full width of the upper housing. The housing or legs shall be so designed that the furnace shall properly stand on a flat surface. Each furnace shall be furnished with two (2) square ceramic trays (3.3.5), a heavy duty power cord (3.3.8), and a pyrometer (3.3.7). The furnaces shall be equipped with an automatic temperature control as specified in 3.3.6.

3.3.1 Housing. The upper and lower sections of the housing shall be fabricated from 0.0478 inch minimum thickness (No. 18 Mfrs. Std. Gage) cold rolled steel.

3.3.2 Door and door frame. The door and door frame shall be fabricated from 0.0598 inch minimum thickness (No. 16 Mfrs. Std. Gage) cold rolled steel.

3.3.3 Heating element. The heating element shall be fabricated from a high oxidation resistant iron-chromium-aluminum alloy wire, having a minimum diameter of 0.045 inch (No. 17 American Wire Gage). The chemical composition shall be 25.5 ± 1 percent chromium (Cr), 5.5 ± 0.6% aluminum (Al), 0.5 to 1.0 percent silicon (Si), 0.1 to 0.2 percent carbon (C) and the balance iron (Fe). The electrical resistivity shall be 875 ± 30 ohms per circular mil foot at 68° Fahrenheit (°F).

3.3.4 Refractories and insulation.

3.3.4.1 Muffle. The molded alumina refractory used for the fabrication of the muffle shall have the following properties:

Aluminum oxide (Al_2O_3) content.	85.5 percent
Silicon oxide (SiO_2) content.	12.1 percent
Modulus of rupture (psi) 1832°.	4500
Linear thermal expansion (in/in/°F) 86 to 2732°F.	4.7×10^{-6}
Thermal conductivity at 1000°F (BTUH/sq ft/in/°F).	10.0
Service temperature (°F maximum) oxidizing atmosphere.	2750

3.3.4.1.1 Muffle insulation. The material used for insulating the muffle shall have the following properties:

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Aluminum oxide (Al_2O_3) content.	38 to 43 percent
Silicon oxide (SiO_2) content.	44 to 52 percent
Modulus of rupture (psi) minimum.	95
Linear thermal expansion (in/in/^oF) maximum 1/.	$3.0x10^{-6}$
Thermal conductivity at 1000^oF (BTUH/sq ft/in/^oF) maximum.	1.20
Service temperature (^oF).	2000

1/ Not applicable to fibrous refractory insulation.

3.3.4.2 Door. The ribbed ceramic facing used on the inside surface of the door shall have the following properties:

Aluminum oxide (Al_2O_3) content.	20 to 23 percent
Silicon oxide (SiO_2) content.	61 to 64 percent
Modulus of rupture (psi) minimum.	3190
Linear thermal expansion (in/in/^oF) maximum.	$3.8x10^{-6}$
Thermal conductivity at 1000^oF (BTUH/sq ft/in/^oF) maximum.	10.0
Service temperature (^oF).	2000

3.3.4.2.1 Door insulation. The insulation material used between the ceramic inner face of the door and the door proper shall have the properties specified in 3.3.4.1.1.

3.3.4.3 Insulation between housing. The insulation material used between the upper and lower sections shall have the properties specified in 3.3.4.1.1.

3.3.5 Ceramic trays. The trays shall be fabricated of molten cast refractory that will withstand the temperatures and atmospheric conditions encountered within the furnace. Each tray shall have a multi-grooved top and a flat bottom with 4 integrally molded legs. The overall height of the trays shall be 15/32 ± 1/32 inches and shall be a minimum of 4 inches square designed to fit on the floor of the muffle side by side when the door is closed.

3.3.6 Automatic temperature control. The furnace shall be equipped with a completely automatic built in temperature controller, by which the desired temperature may be preset upon starting the furnace. The preset temperature shall be reached with full power input to the furnace. Once it is reached, the desired temperature shall be maintained regardless of normal line voltage fluctuation and changes in heating element resistance. Any temperature within the range of the furnace shall be accurately set and maintained within ± 50^oF (± 28^oC). The automatic control

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system shall be rated for at least 125 percent of the maximum current drawn at 115 volts. An equilibrium temperature of 1700° ± 50° F (926° ±28° C) shall be attained within 3 hours when tested as specified in 4.4.7.

3.3.7 Pyrometer. The furnace shall be equipped with a pyrometer for indicating the furnace temperature. The pyrometer scale shall be calibrated over a minimum range of 250° F to 2000° F in increments of 50° F or less. The scale shall not be less than 1-1/2 inches in overall length.

3.3.7.1 Thermocouple. The pyrometer shall be equipped with an 0.080 inch diameter (No. 14 Steel Wire Gage) minimum, Chromel/Alumel (type K) thermocouple. The thermocouple shall extend a minimum of 3/4 inch into the furnace area.

3.3.7.2 Thermocouple failure protection. The furnace shall be equipped with a thermocouple failure protection circuit that automatically turns off power to the furnace if the thermocouple fails.

3.3.8 Power cord. The power cord shall be a three conductor, 16 AWG minimum, type HS or HSJ cable terminating in a 3-blade male plug suitable for use with receptacles specified in ANSI C73.11. A strain relief bushing shall be provided where the power cord enters the housing. The cord shall have a minimum external length of six feet. The terminal connections inside the housing shall show no evidence of strain when tested as specified in 4.4.3. The ground lead shall be so connected that all exposed metal parts of the furnace are electrically grounded when tested as specified in 4.4.4.

3.3.9 Dimensions.

3.3.9.1 Heating chamber. The heating chamber shall have the following dimensions:

Width	9 ± 1/4 inches
Height	4 ± 1/4 inches
Working depth	5 ± 1/4 inches

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3.3.9.2 Overall dimensions. The overall dimensions of the furnace excluding counterweight, with the door closed shall not exceed the following:

Width	17 inches
Height	17 1/2 inches
Depth	11 1/2 inches

3.3.10 Finish. All metal surfaces of the furnace housing and door, inside and outside, shall be finished to prevent corrosion. All outside metal surfaces shall be painted with a durable heat-resistant enamel using commercial painting processes. The color shall be black crinkle finish. The control panel shall have a smooth black finish with symbolization in contrasting colors.

3.4 Performance.

3.4.1 Calibration. The pyrometer indicated temperature shall be accurate within ± 50° F (± 28° C) at 1300° F (704° C) when tested as specified in 4.4.5.

3.4.2 Heating rate. The time required to increase the temperature of the furnace from room temperature to 1600° F (871° C) shall not exceed 75 minutes when tested as specified in 4.4.6.

3.4.3 Power rating. The maximum power consumed by the furnace shall be 1650 watts when tested as specified in 4.4.8.

3.4.4 Dielectric withstand. There shall be no electrical breakdown when tested as specified in 4.4.9.

3.4.5 Leakage current. The leakage current shall be not more than 0.5 milliamperes when tested as specified in 4.4.10.

3.5 Identification marking. Each furnace shall have a nameplate of corrosion-resisting metal, securely mounted in a conspicuous location, and permanently and legibly marked with the following information:

(a) Manufacturer's name or registered trademark.
(b) Model number.
(c) Voltage.
(d) Frequency.
(e) Power requirements.

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3.6 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.7 Workmanship. Workmanship shall be first class throughout. The furnace shall be free from defects which detract from their appearance, or impair their serviceability, or may endanger the safety of the operator. All metal components shall be free of burrs, sharp edges and deformities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one furnace.

TABLE I. Sampling for examination.

Categories	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for test purposes shall be one furnace.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Chemical analysis (insulation and refractories).	3.3.4.1 3.3.4.1.1 3.3.4.2 3.3.4.2.1 3.3.4.3	ASTM-C573	1/	1/
Chemical analysis (heating element wire).	3.3.3	4.4.1	1/	1/

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TABLE II. Sampling for tests. -(Continued)

Characteristic	Requirement	Test Inspection	Inspection level	AQL (percent defective)
Modulus of rupture.	3.3.4.1 3.3.4.2	ASTM-C674	S-1	4.0
Modulus of rupture.	3.3.4.1.1 3.3.4.2.1 3.3.4.3	ASTM-C93	S-1	4.0
Linear thermal expansion.	3.3.4.1 3.3.4.1.1 3.3.4.2 3.3.4.2.1 3.3.4.3	ASTM-C372	S-1	4.0
Thermal conductivity.	3.3.4.1	ASTM-C202	S-1	4.0
Thermal conductivity.	3.3.4.1.1 3.3.4.2.1 3.3.4.3	ASTM-C182	S-1	4.0
Thermal conductivity.	3.3.4.2	ASTM-C408	S-1	4.0
Electrical resistivity.	3.3.3	ASTM-B193	S-1	4.0
Door counterbalance test.	3.3	4.4.2	S-1	4.0
Strain relief.	3.3.8	4.4.3	S-1	4.0
Continuity to ground.	3.3.8	4.4.4	100%	4.0
Calibration.	3.4.1	4.4.5	S-1	4.0
Heating rate.	3.4.2	4.4.6	S-1	4.0
Equilibrium temperature.	3.3.6	4.4.7	S-1	4.0
Power rating.	3.4.3	4.4.8	S-1	4.0
Dielectric withstand.	3.4.4	4.4.9	100%	4.0
Leakage current.	3.4.5	4.4.10	100%	4.0

1/ Failure to meet requirements specified shall be cause for rejection of the lot.

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4.3 Examination.

4.3.1 Furnace. The furnace shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in table I and table III. Examination however, shall not be restricted to defects listed in table III.

TABLE III. Classification of defects.

Classification	Defect
Major;	
101	Muffle cracked.
102	Insulation missing around muffle.
103	Refractory in door cracked.
104	Insulation between door refractory and door proper missing.
105	Insulation between housing sections missing.
106	Parts or accessories missing.
107	Metal parts not free of corrosion.
108	Metal parts not free of burrs or sharp edges.
109	Door not adequately attached to housing.
110	Door not capable of being easily opened or closed.
111	Door does not close tightly.
112	Internal wiring not adequately secured to terminations.
113	Controls do not operate smoothly.
Minor;	
201	Finish not free of scratches, chipping, cracking or peeling.
202	Identification marking not complete, correct, legible or permanent.
203	Unit not free of foreign matter.

4.3.2 Dimensional examination. The furnace shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

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TABLE IV. Examination of preparation for delivery.

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packing and packaging material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-566, PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636 or PPP-B-676. The examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis. A sufficient quantity of each lot of metal used in the manufacture of the heating element wire shall be selected for chemical analysis in accordance with Fed. Test Method Std. No. 151, Methods 111 or 112.

4.4.2 Door counterbalance test. The counterbalance test shall be performed at the two extreme positions of the door, i.e. fully open and fully closed. With the door in the fully closed position, a spring balance of adequate capacity, graduated in two ounce intervals, shall be attached by a suitable means to the longitudinal center of the counterweight. A force shall be applied perpendicular to the longitudinal axis in the direction to open the furnace door. The force shall be slowly and continuously applied perpendicular to the longitudinal axis until the door is fully open. The maximum force required to completely open the door shall be recorded. The test as specified above shall be repeated in the opposite direction and the maximum force to fully close the door shall be recorded.

4.4.3 Strain relief. The power cord strain test shall be conducted in accordance with UL-1262.

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4.4.4 Continuity to ground. A continuity indicating device shall be used to check continuity between the ground pin of the input plug at the end of the power cord and all exposed metal parts of the furnace. Grounding procedures shall be in accordance with UL-1262.

4.4.5 Calibration. Each furnace shall be tested for calibration accuracy by the use of Tempil pellets. A Tempil pellet compounded to melt at 1300° F (704° C) shall be used. The following procedure shall be followed:

(a) Place an empty medium size flask (inlay casting ring) approximately at the center of the muffle chamber.

(b) Set a small, thin, metal plate or tray on top of the flask and place a 1300° F (704° C) Tempil pellet at the center of the tray.

(c) Close the door of the furnace and turn the power switch on.

(d) Set the temperature set knob at 1000° F (538° C).

(e) After the temperature indicator has risen to 1000° F (538° C), heat soak for three minutes.

(f) After the three minute 1000° F (538° C) soak, set the temperature indicator to 1500° F (816° C).

(g) When the temperature indicator shows 1150° F (621° C) begin to check for melt every 25° F. When checking, open the furnace door just enough to determine at a quick glance if the Tempil has begun to liquify around the edges. Keep the furnace door closed as much as possible during this observation period to prevent heat loss.

(h) When the Tempil pellet begins to melt or liquify around the edges the temperature indicator shall show 1300° F (704° C). A 1300° F Tempil pellet has a melting point of 1300° ± 10° F (704° ± 5.5° C).

4.4.6 Heating rate. The furnace shall be connected to a regulated 115V., 60 Hz supply. With the temperature controller set for 1600° F (871° C) the time required to reach 1600° F (871° C) from a room ambient of 68° F (20° C) shall be as specified in 3.4.2.

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4.4.7 Equilibrium temperature. The furnace shall be connected to a regulated 115V, 60 Hz supply. The furnace shall be empty and the door closed. The temperature controller shall be set at 1700° F (926° C). Equilibrium shall be attained as specified in 3.3.6 when test is performed at room ambient 68° F (20° C).

4.4.8 Power rating. The power consumed shall be determined by a suitable watt meter with the furnace operating at rated voltage and at the equilibrium temperature attained in 4.4.7.

4.4.9 Dielectric withstand. The dielectric withstand test shall be performed in accordance with UL-1262.

4.4.10 Leakage current. The leakage current test shall be performed in accordance with UL-1262.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each furnace shall be packaged in a box conforming to PPP-B-636, class domestic. The furnace shall be cushioned within the box with corrugated fiberboard pads or other suitable material. Closure of the box shall be as specified in the appendix of the box specification.

5.1.2 Level C. Each furnace shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment form the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each packaged furnace shall be packed in an exterior container designed for a type 1 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; or PPP-B-621, class 2. The box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing of the case liner shall be as specified in the appendix of MIL-L-10547. Closure and strapping of the box shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Each packaged furnace shall be packed in an exterior container designed for a type 1 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; or PPP-B-621, class 1. Closure shall be as specified in the appendix of the appliable box specification.

5.2.3 Level C. Each packaged furnace shall be packed in a shipping container that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

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5.3 Marking.

5.3.1 Civil agencies. In addition to any marking specified in the contract or order, packages and shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123.

5.3.2 Military agencies. Interior packages and exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The furnace covered by this specification is intended for burn-out and heat treating performed in Federal Government dental laboratories.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) National stock number (NSN).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable marking (see 5.3).

6.3 This specification is intended to cover the following item:

National stock number	Item Identification
6520-00-966-3729	Furnace, Dental Laboratory, Electric, Muffle Type, Large with Pyrometer, Automatic Temperature Control, 115 volt, 60 Hertz.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Civil Agencies Coordinating Activities:
GSA-FSS
COM-NBS

Project No. 6520-1444

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies of documents referenced herein. Price 1.00cents each.

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W-F-779D FURNACE, DENTAL LABORATORY, ELECTRIC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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